

Python

Complete Course

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Course Roadmap

Slide Content

Welcome to the Python Complete Course!

This course is carefully designed to take you from a complete beginner to an advanced Python programmer. Throughout 20 weeks, you will learn Python fundamentals, object-oriented programming, file handling, advanced Python concepts, and practical libraries like NumPy and Pandas. By the end of this course, you will have the knowledge and confidence to build real-world Python applications and continue your journey into AI, automation, data science, or web development.





Course Roadmap

Weeks 1–7

- Python Fundamentals
- Variables & Data Types
- Operators
- Strings
- Conditional Statements
- Loops
- Functions

Weeks 8–14

- Lists & Tuples
- Sets & Dictionaries
- Classes & Objects
- Inheritance
- Encapsulation
- Polymorphism
- Abstraction

Weeks 15–20

- File Handling
- Exception Handling
- Modules & Packages
- Advanced Python
- NumPy & Pandas
- Final Projects



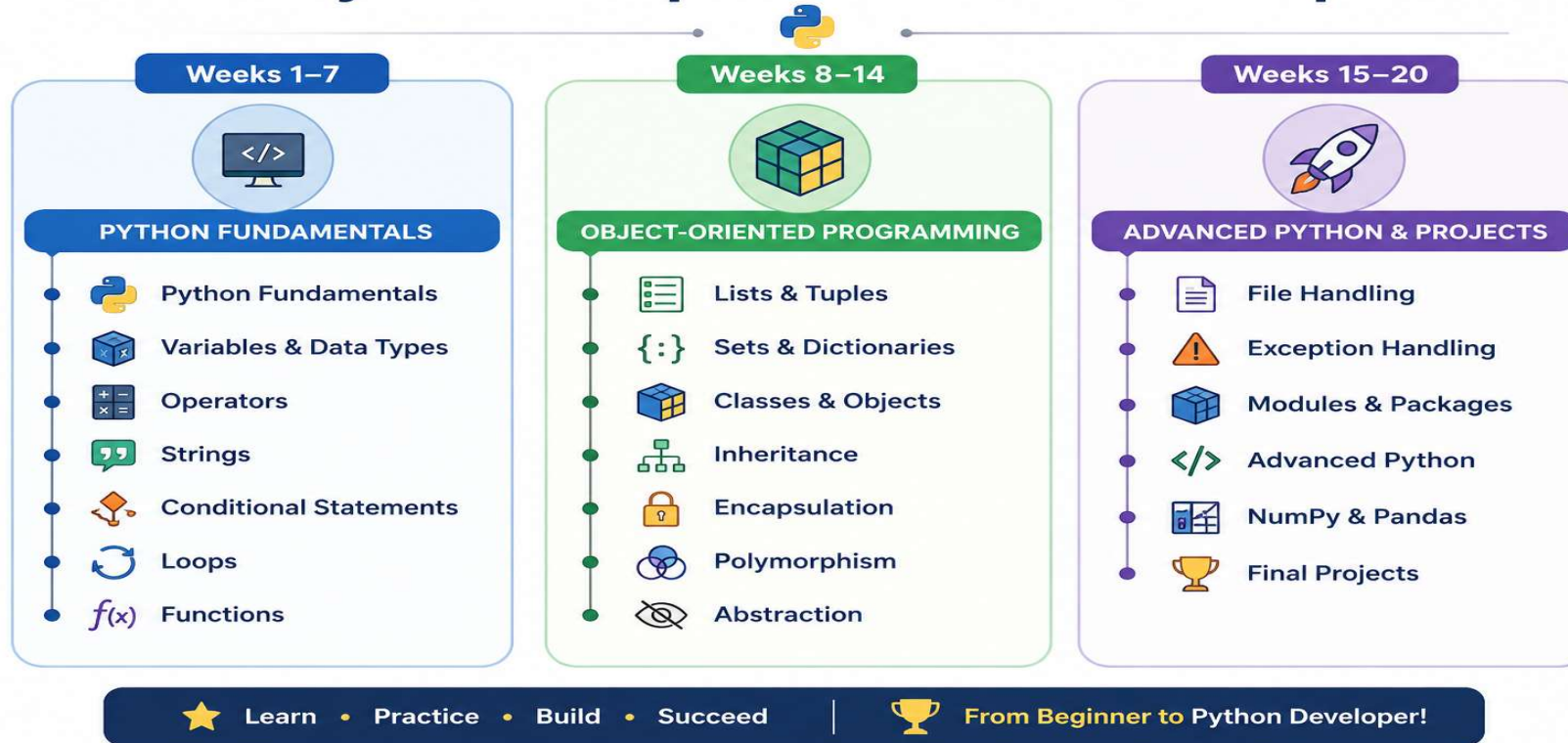
Tip:

Master each week's concepts before moving to the next. Python is best learned through consistent practice.



 Course Roadmap

Python Complete Course Roadmap



Week 1 Overview

Week 1: Introduction to Python

Learning Objectives

By the end of this week, you will be able to:

-  Explain what Python is.
-  Describe the history of Python.
-  Identify the key features of Python.
-  Explain why Python is popular.
-  Describe the major applications of Python.
-  Install Python on your computer.
-  Install and configure Visual Studio Code.
-  Create and run your first Python program.
-  Understand Python syntax and indentation.
-  Write comments in Python programs.
-  Recognize Python keywords and identifiers.
-  Follow basic Python coding standards.

Skills You Will Gain

-  Python Fundamentals
-  Programming Basics
-  Environment Setup
-  Writing Your First Program
-  Problem-Solving Mindset



Week 1 Topics

- 1.What is Python?
- 2.Why Learn Python?
- 3.History of Python
- 4.Features of Python
- 5.Applications of Python
- 6.Companies Using Python
- 7.Installing Python
- 8.Installing VS Code
- 9.Running Your First Program
- 10.Python Syntax
- 11.Indentation
- 12.Comments
- 13.Keywords
- 14.Identifiers
- 15.Best Practices



What is Python?

Python is a high-level, interpreted, general-purpose programming language.

Easy to read, write, and understand. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming, making it suitable for beginners as well as professional software developers.

Simple Explanation

Think of Python as a language that allows humans to communicate with computers using simple and readable instructions. Unlike many other programming languages that require complex syntax, Python focuses on clarity and simplicity, enabling programmers to solve problems with fewer lines of code. This is one of the main reasons it is widely used in education, industry, and research.



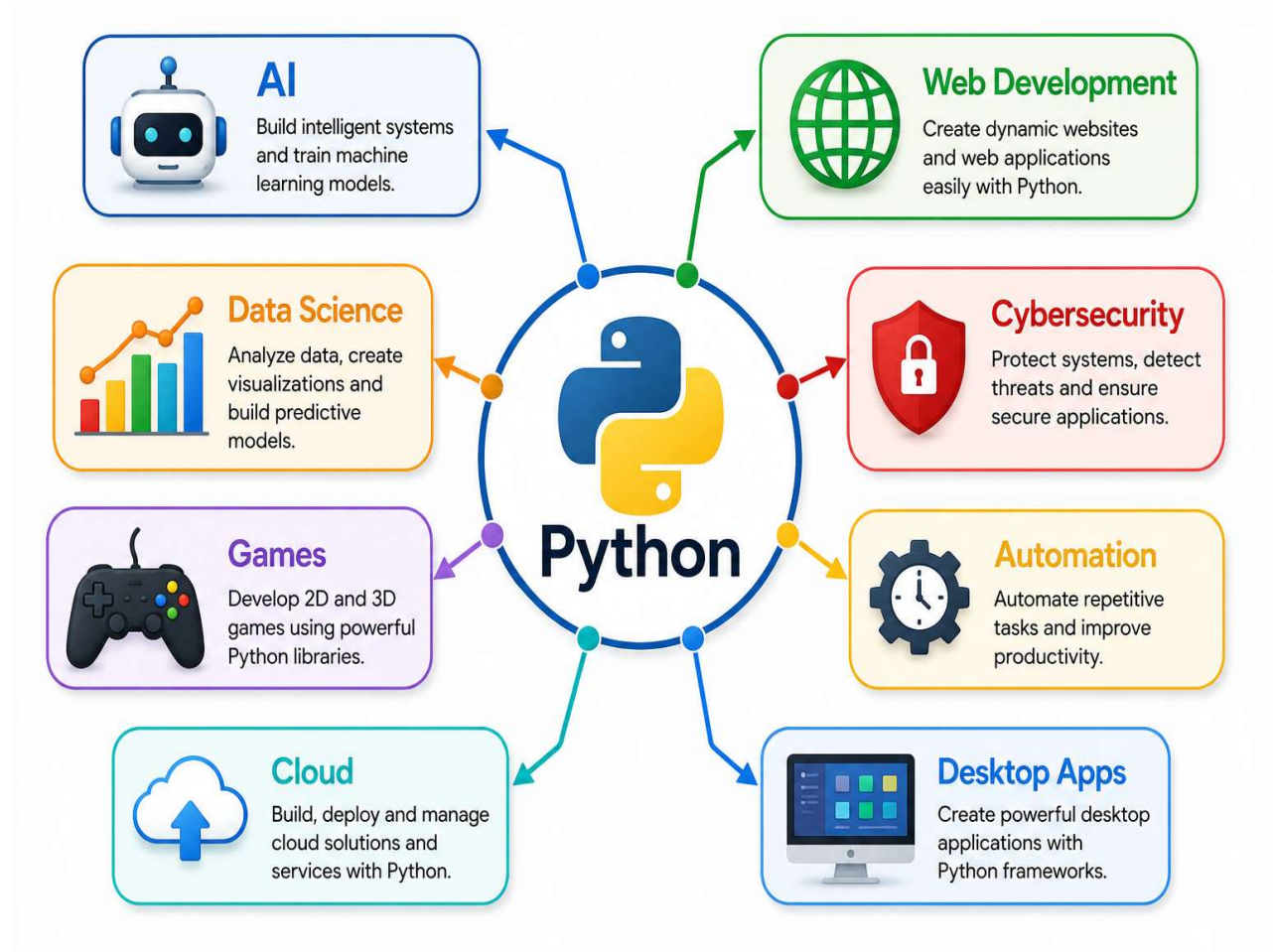
★ Why Learn Python?

- ✓ Easy to Learn and Read
- ✓ Beginner-Friendly Syntax
- ✓ High Demand in the Job Market
- ✓ Used in Artificial Intelligence & Machine Learning
- ✓ Popular for Data Science & Data Analysis
- ✓ Powerful for Automation and Scripting
- ✓ Excellent for Web Development
- ✓ Large Collection of Libraries and Frameworks
- ✓ Strong Community Support
- ✓ Free and Open Source
- ✓ Cross-Platform Compatibility
- ✓ Suitable for Beginners and Professionals



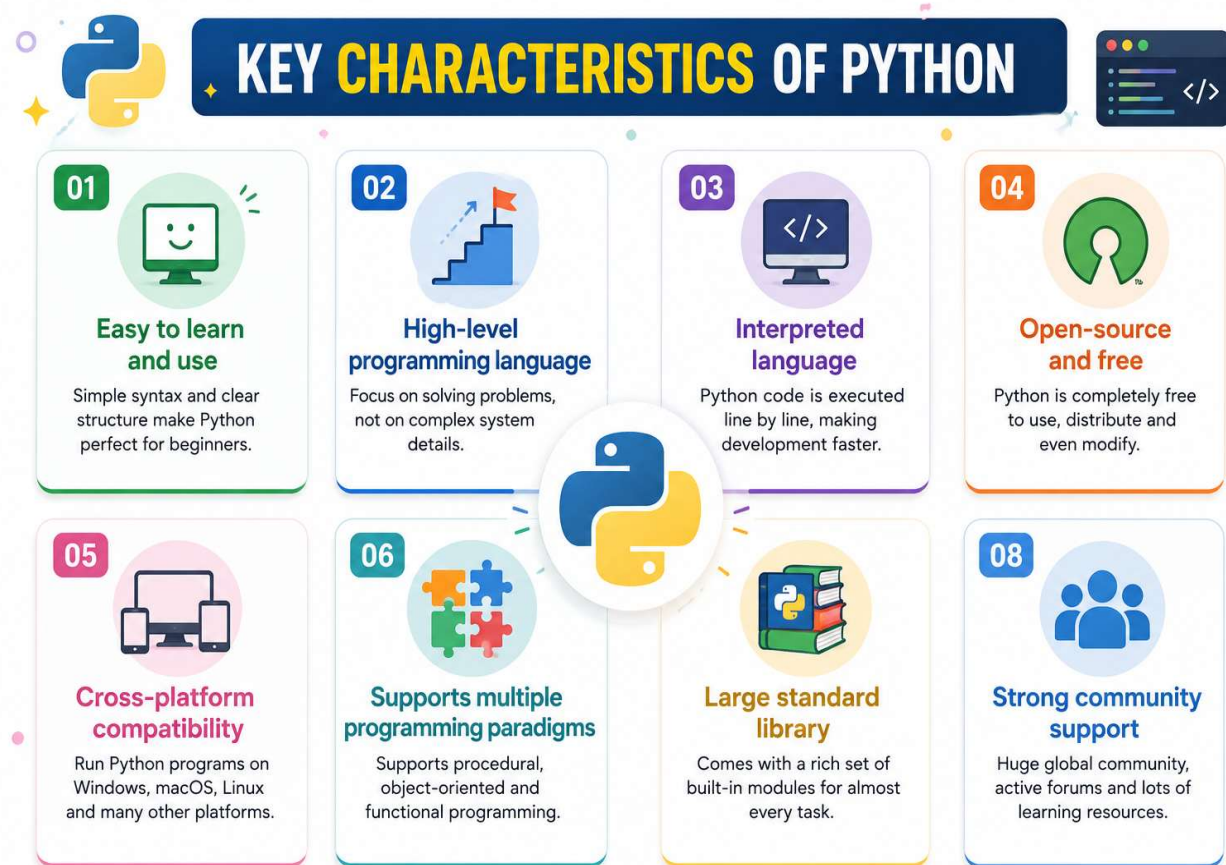
Where is Python Used?

- Artificial Intelligence
- Machine Learning
- Data Science
- Web Development
- Game Development
- Cybersecurity
- Automation
- Cloud Computing
- Desktop Applications
- Scientific Computing



★ Key Characteristics

- Easy to learn and use
- High-level programming language
- Interpreted language
- Open-source and free
- Cross-platform compatibility
- Supports multiple programming paradigms
- Large standard library
- Strong community support



History of Python

Introduction

Python was created by **Guido van Rossum** in the late 1980s and officially released in **1991**. The goal was to develop a programming language that was simple, readable, and powerful enough to solve real-world problems. Over the years, Python has evolved into one of the most widely used programming languages, powering applications in artificial intelligence, web development, data science, automation, cybersecurity, scientific research, and many other fields.

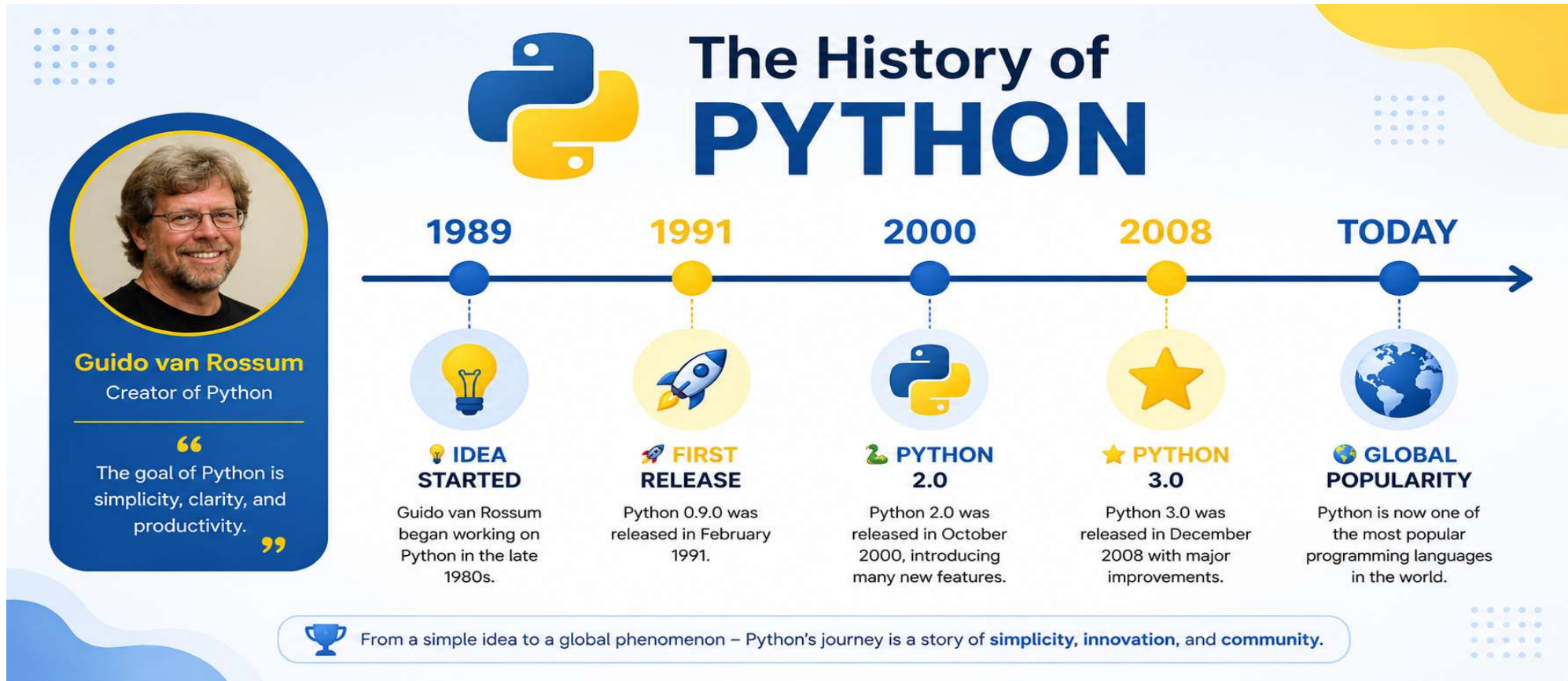


Creator of Python

- **Name:** Guido van Rossum
- **Nationality:** Dutch
- **Started Development:** 1989
- **First Public Release:** 1991
- **Known As:** "Benevolent Dictator For Life (BDFL)" (historical title used during Python's early development)



History of Python



Features of Python

★ Key Features of Python

Feature

Simple & Easy to Learn

High-Level Language

Interpreted Language

Cross-Platform

Open Source

Object-Oriented

Large Standard Library

Large Community

Description

Python has a clean and readable syntax, making it ideal for beginners.

Developers focus on solving problems instead of managing hardware details.

Python executes code line by line without the need for compilation.

Python programs run on Windows, macOS, and Linux with little or no modification.

Python is free to use, modify, and distribute.

Supports classes, objects, inheritance, encapsulation, and polymorphism.

Includes built-in modules for file handling, networking, mathematics, and more.

Millions of developers contribute tutorials, libraries, and community support.



Applications of Python

Application

-  Artificial Intelligence
-  Machine Learning
-  Data Science & Analytics
-  Web Development
-  Automation & Scripting
-  Cloud Computing
-  Cybersecurity
-  Game Development
-  Desktop Applications
-  Scientific Computing

Example

Chatbots, recommendation systems, image recognition

Predictive models, fraud detection, speech recognition

Data visualization, business intelligence, statistics

Websites and web applications

Automating repetitive tasks

Cloud services and deployment

Security tools, penetration testing, log analysis

2D and 3D games

GUI software using PyQt, Tkinter, PySide

Research, simulations, numerical analysis





Top Companies Using Python

Company

How Python is Used

Google

Search systems, AI research, automation

Microsoft

AI, cloud services, automation

Netflix

Recommendation engine, data analysis

Instagram

Backend web development

Spotify

Music recommendation system

NASA

Scientific computing and research

OpenAI

Artificial intelligence and machine learning

Dropbox

Cloud storage services

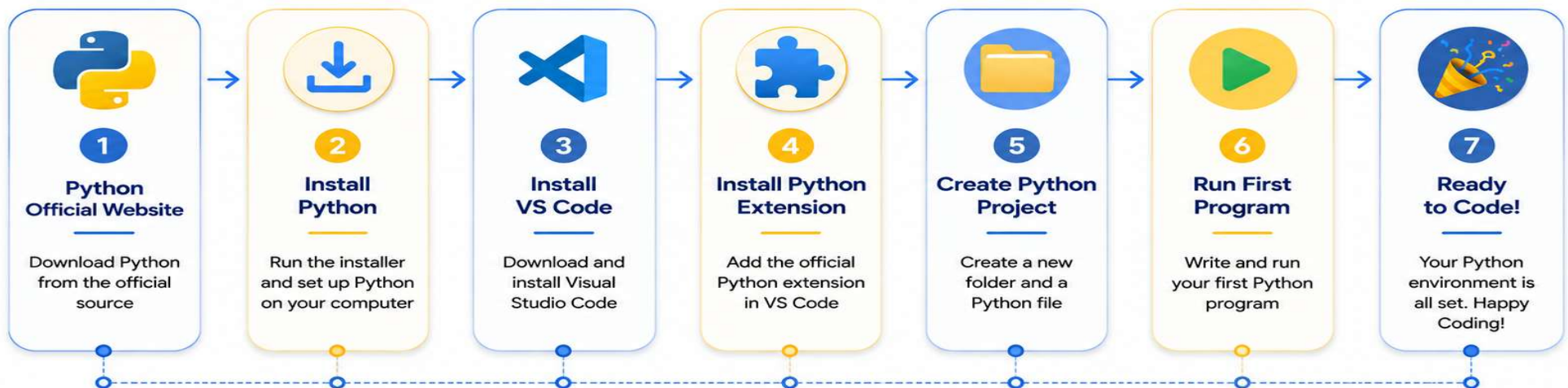
Reddit

Website backend and automation



Setting Up the Python Environment

Setting Up the Python Environment



A properly configured environment helps you write, run, and manage Python programs efficiently.



Downloading & Installing Python

Installation Steps

Step 1

Open your preferred web browser (Google Chrome, Microsoft Edge, etc.).

Step 2

Visit the official Python website:

<https://www.python.org>

Step 3

Click the **Downloads** menu.

Step 4

The website automatically detects your operating system.

You will see a button similar to:

Download Python 3.x.x

Click the download button.

Step 5

Wait for the installer (.exe on Windows) to finish downloading.



Download Python

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Download Python 3.12.3

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2 Python » Downloads

Download Python

Download Python 3.12.3

Looking for Python with a different OS? [Windows](#), [Linux/UNIX](#), [macOS](#), [Other](#)

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3

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27.6 MB • Done

Full download history

4

Quick access

This PC 3D Objects Desktop Documents Downloads Pictures Videos

Today (1)

python-3.12.3-amd64.exe

Downloads Page **Browser Download** **Downloaded Installer**

Learning Objective

By the end of this lesson, you will be able to download the latest stable version of Python from the official website.

- 1** **Open Your Web Browser**
Open any modern web browser (Chrome, Edge, Firefox, etc.).
- 2** **Visit the Official Python Website**
Go to <https://www.python.org>
Always download Python from the official website.
- 3** **Click the Downloads Menu**
Click on the "Downloads" menu in the top navigation bar.
- 4** **Download the Latest Version**
Click the yellow "Download Python 3.x.x" button.
The download will start automatically.
- 5** **Wait for the Download to Complete**
Once downloaded, you will get the installer file (.exe) in your Downloads folder.

Tip Download the latest stable release unless your instructor or project requires a different version.

Common Mistake

Downloading Python from unofficial websites.
This may result in outdated or unsafe installers.

Key Takeaway

You have successfully downloaded the official Python installer and are now ready for the installation process.



Install Python

1 Run the Installer

Double-click the downloaded installer file (.exe) to start the installation.



2 Check "Add Python to PATH"

Make sure to check the box "Add Python to PATH" before clicking Install Now.



3 Installation Progress

Wait while Python is being installed on your computer.



4 Installation Successful

You will see the success message. Click Close to finish.



Learning Objective

By the end of this lesson, you will be able to install Python correctly and add it to your system PATH.

Step-by-Step Guide

1 Run the Installer

Open the downloaded .exe file from your Downloads folder.

2 Check "Add Python to PATH" (Very Important)

This option allows you to run Python from anywhere in Command Prompt.

3 Click Install Now

Leave the default settings and click Install Now.

4 Wait for Installation

The installation process may take a few minutes.

5 Installation Successful

Click Close. Python is now installed on your computer.



Tip

We recommend leaving the default installation settings for most users.



Common Mistake

Forgetting to check "Add Python to PATH". This may cause 'python is not recognized' error in Command Prompt.



Key Takeaway

You have successfully installed Python on your system. Now you are ready to verify the installation.



Supported Platforms:



Windows



macOS



Linux



Always download Python from the official website: <https://www.python.org>



Download Visual Studio Code

Step-by-Step Guide

Step 1

Open your preferred web browser

Step 2

Visit the official VS Code website:

<https://code.visualstudio.com>

Step 3

Click the **Download for Windows** button.

(On macOS or Linux, download the version for your operating system.)

Step 4

Wait for the installer to download completely.

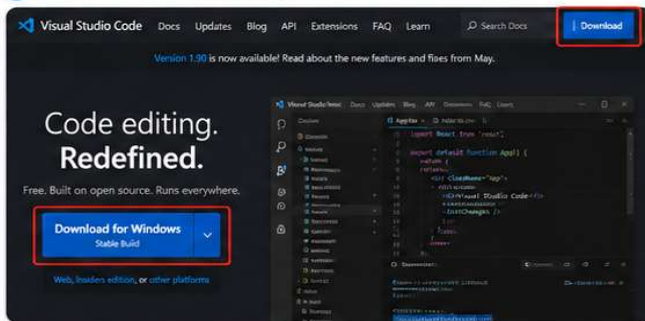
The downloaded file will be similar to:

Step 5

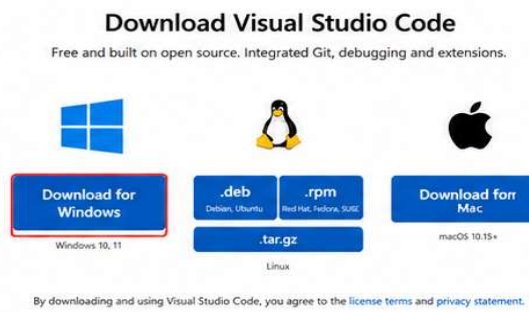
Open your **Downloads** folder and verify that the installer has been downloaded successfully

Download Visual Studio Code

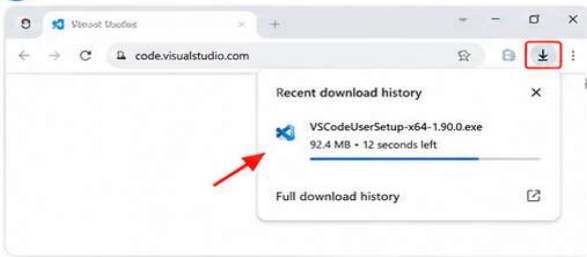
1 VS Code Official Homepage



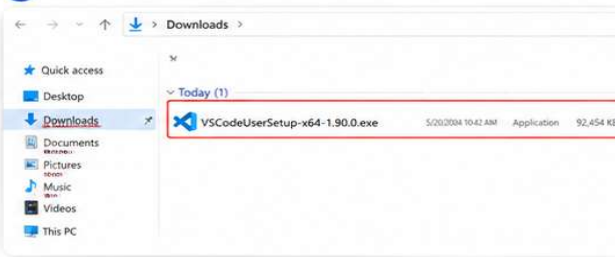
2 Download Page



3 Browser Download



4 Downloaded Installer



Supported
Platforms



Windows
Windows 10, 11
(64-bit)



macOS
macOS 10.15+
(64-bit)



Linux
Debian, Ubuntu, Red Hat,
Fedora, SUSE (64-bit)

Learning Objectives

- Visit the official VS Code website.
- Download the latest version of Visual Studio Code.
- Select the correct installer for your operating system.

Step-by-Step Guide

- 1 Open Your Browser**
Open Google Chrome, Microsoft Edge, Firefox, or any browser.
- 2 Visit the Official Website**
Go to <https://code.visualstudio.com>
- 3 Click Download for Windows**
Click the "Download for Windows" button (choose your OS if macOS or Linux).
- 4 Wait for the Download**
The installer will start downloading.
- 5 Open Downloads Folder**
Go to your Downloads folder and verify that the installer file is downloaded.



Tip
Always download VS Code from the official Microsoft website to get the latest and most secure version.



Common Mistake
Downloading from third-party websites may give you outdated or even harmful software.

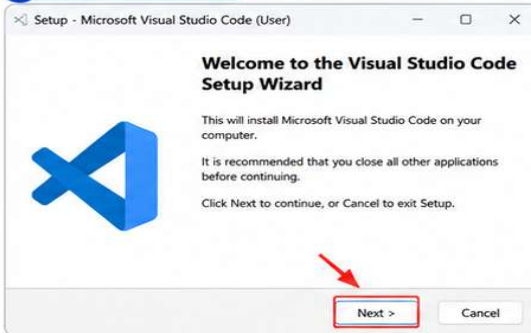


Key Takeaway
Downloading Visual Studio Code from the official website is the first step toward a powerful and reliable coding setup.

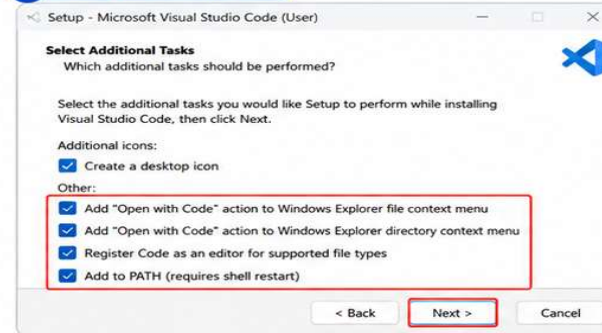


Install Visual Studio Code

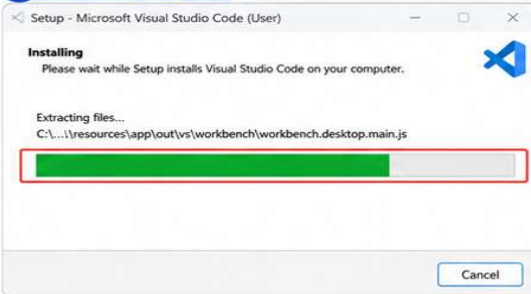
1 Installer Welcome



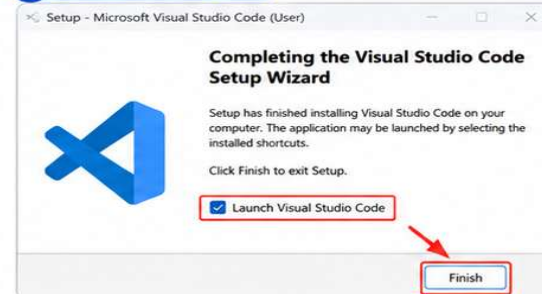
2 Select Additional Tasks



3 Installing Progress



4 Installation Complete



 **Visual Studio Code
is now installed!**

 **Lightweight
Fast & Powerful**

 **Extensible with
Many Extensions**

 **Perfect for
Python Development**

Learning Objectives

- Install Visual Studio Code successfully.
- Choose the recommended installation options.
- Configure VS Code for Python development.
- Launch VS Code after installation.

Step-by-Step Installation

- 1 Run the Installer**
Double-click the downloaded installer.
- 2 Accept License Agreement**
Select "I accept the agreement" and click Next.
- 3 Choose Installation Location**
Keep the default location and click Next.
- 4 Recommended Options** ★
Select the recommended options (shown on the left) and click Next.
- 5 Install**
Click Install and wait for the installation to complete.
- 6 Finish**
Click Finish and keep "Launch Visual Studio Code" checked.

 **Tip**
Selecting the recommended options during installation saves time and makes VS Code easier to use.

 **Common Mistake**
Skipping the "Open with Code" or "Register Code as an editor" options. These features make opening projects much faster.

 **Key Takeaway**
A correct VS Code installation ensures a smooth development experience and prepares your computer for Python programming.

Install the Python Extension in VS Code



Step-by-Step Guide

Step 1

Open **Visual Studio Code**.

Step 2

Click the **Extensions** icon on the left sidebar.

(Shortcut: **Ctrl + Shift + X**)

Step 3

In the search box, type:

Python

Step 4 ★ Most Important

Select the extension:

•**Name:** Python

•**Publisher:** Microsoft

Verify that the publisher is **Microsoft** before installing.

Step 5

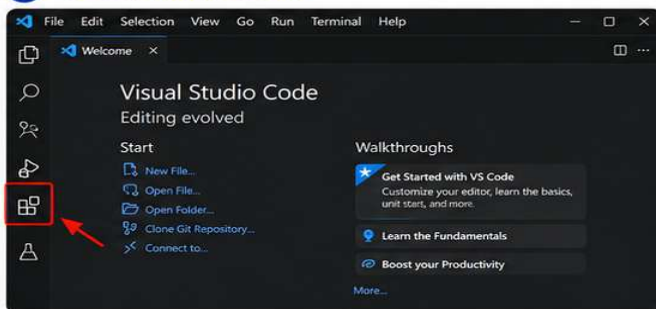
Click the **Install** button.

Wait a few moments for the installation to complete.

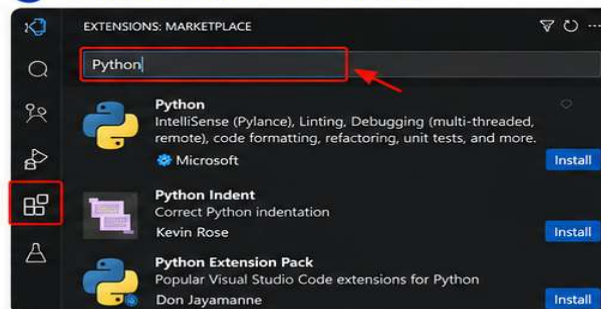


Install Python Extension in VS Code

1 VS Code Home Screen



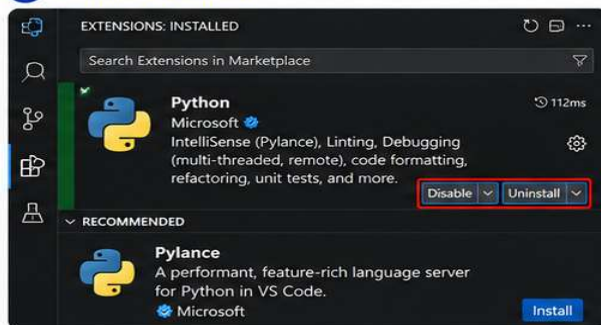
2 Extensions Panel (Search: Python)



3 Python Extension Page (Install)



4 Installed Extension



Extensions make VS Code powerful and customizable.



The Python extension adds intelligence to your coding.



You are now ready to write, run, and debug Python code!

Learning Objectives

- Understand what VS Code extensions are.
- Install the official Python extension.
- Identify the publisher (Microsoft).
- Prepare VS Code for Python development.

Step-by-Step Guide

- 1 Open VS Code**
Launch Visual Studio Code on your computer.
- 2 Open Extensions**
Click the **Extensions** icon on the left sidebar (Shortcut: **Ctrl + Shift + X**).
- 3 Search for Python**
Type **Python** in the search box.
- 4 Select Python by Microsoft**
Choose the extension published by Microsoft.
Always verify the publisher is Microsoft.
- 5 Click Install**
Click the **Install** button and wait for the installation.
- 6 Reload / Restart**
If prompted, reload or restart VS Code to activate the extension.

★ Features You'll Get

- | | |
|--|---------------------------|
| ✓ IntelliSense (Smart Code Completion) | ✓ Auto Import Suggestions |
| ✓ Syntax Highlighting | ✓ Code Formatting |
| ✓ Debugging Support | ✓ Run Python Files |
| ✓ Code Linting | ✓ Testing Support |



Tip

Always install the official Python extension published by Microsoft for the best compatibility and support.



Common Mistake

Installing a third-party Python extension with a similar name. Check that the Publisher is **Microsoft**.



Key Takeaway

The official Python extension transforms VS Code into a powerful Python development environment with intelligent coding assistance and debugging tools.



Create Your First Python Project

Step-by-Step Guide

Step 1 — Create a Folder

Create a new folder on your computer.

Example:

Python-Course

Step 2 — Open VS Code

Launch **Visual Studio Code**.

Step 3 — Open the Folder

Click:

File → Open Folder

Select the **Python-Course** folder.

Step 4 — Create a Python File

In the Explorer panel, click **New File**.

Name the file:

main.py

Step 5 — Write Your First Code

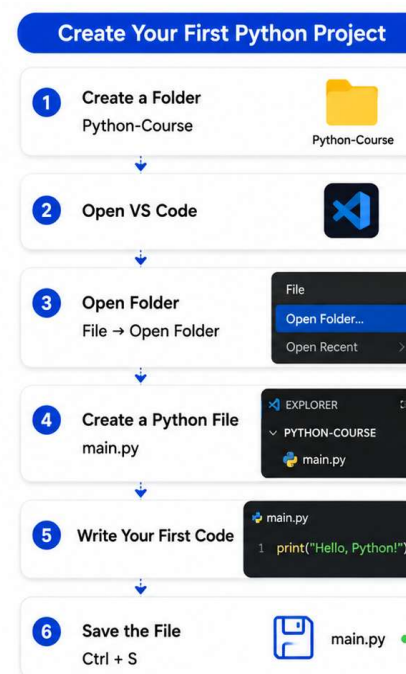
Type:

```
print("Hello, Python!")
```

Step 6 — Save the File

Press:

Ctrl + S



Your First Python Program

```
print("Hello, Python!")
```

